

Customer Churn Analysis Report

This report presents a comprehensive customer churn analysis using Power BI, DAX, and data cleaning techniques. The objective is to identify churn drivers, quantify revenue impact, and provide actionable recommendations for improving customer retention and business performance.

Project Overview

This dashboard analyzes customer churn behavior and identifies key drivers behind customer attrition. The project focuses on improving retention through data-driven insights. Key KPIs such as total customers, churn rate, and retention metrics are highlighted. The analysis supports strategic decision-making for customer retention initiatives.

Customer Churn Analysis Dashboard

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Problem Statement:

The objective of this project is to analyze the customer churn behavior and identify the key factors contributing to customer attrition. By leveraging data visualization techniques, this dashboard aims to help businesses understand churn patterns and take proactive actions to improve customer retention.

Dashboard Created:

1.Customer Churn Analysis & Retention

Insights - Worked on KPIs like Total Customers, Churn Rate, Retention Rate, Revenue loss and gathered insights like ~**18%** revenue loss due to Contract type and Partner Status and recommending to reduce monthly charges and strategies for retention improvement.

2.Customer Segmentation & Churn

Behavior Insights - Worked on KPIs like Churn rate, Total tenure, Avg. monthly charges which leads to insights that ~**2M** Revenue loss due to monthly plans, recommending to reduce the monthly charges for increasing the total tenure on Monthly Basis.

Key Metrics Preview

- 1.Total Customers
- 2.Churned Customers
- 3.Churn Rate %

Tool Used

- 1.Power Bi
- 2.DAX
- 3.Data Cleaning

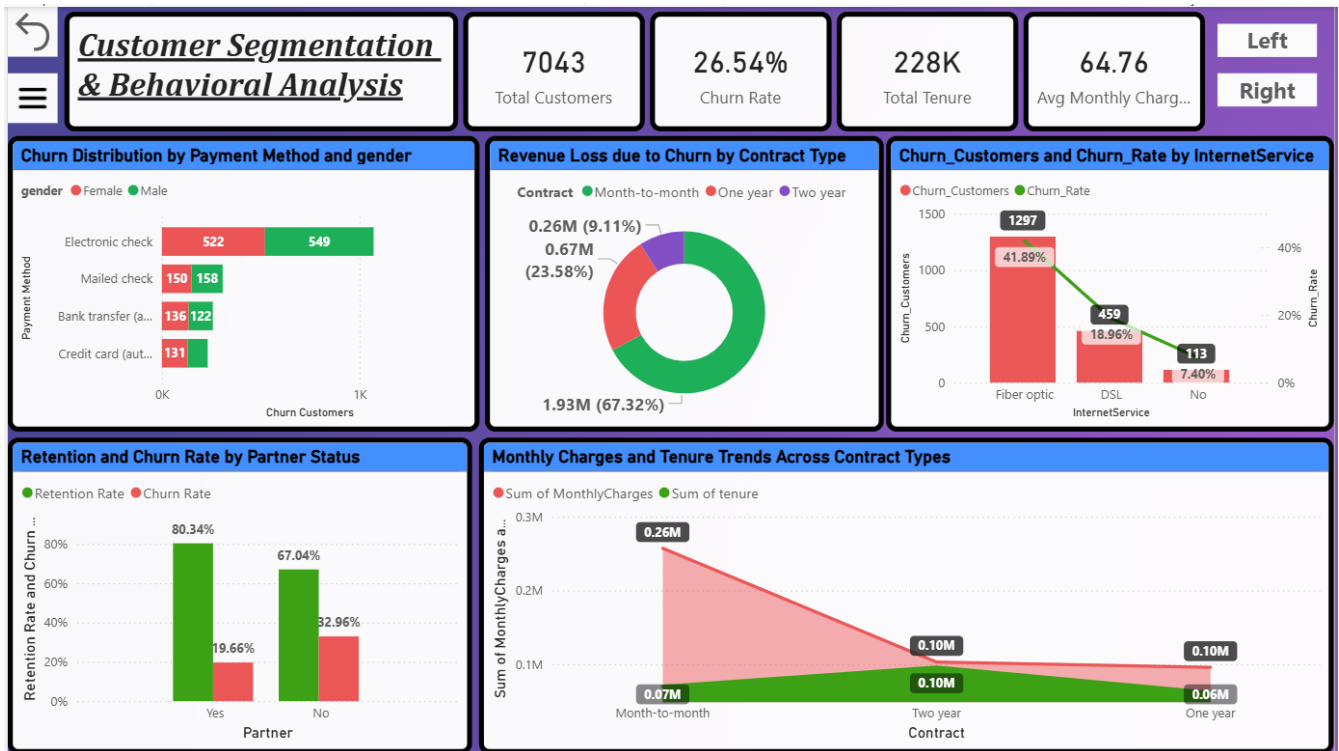
Churn & Retention Insights

Month-to-month contracts exhibit the highest churn rate, indicating low customer commitment. Revenue loss is significantly concentrated within high-risk customer segments. Payment methods and device protection services influence retention outcomes. These insights help prioritize retention-focused business strategies.



Customer Segmentation Analysis

Customer behavior is segmented by payment method, gender, internet service, and partner status. Fiber optic users show the highest churn rates despite contributing substantial revenue. Customers without partners demonstrate a greater likelihood of attrition. The segmentation analysis enables targeted engagement and retention campaigns.



Key Business Insights

Contract type remains the strongest predictor of customer churn. Electronic check users are more likely to discontinue services than auto-payment users. Value-added services such as device protection improve customer retention rates. Higher monthly charges combined with lower tenure increase churn probability.

Key Insights:

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1.Contract Type Insights:

- 1.Month-to-Month contracts contributing to highest customer churn (~42.7%), indicating low customer commitment and higher risk of revenue loss.

2.Revenue Type Insights:

- 1.Approximately ~67% of total revenue loss is driven by month-to-month customers, making them the most critical segment for retention strategies.
- 2.Over all approximately ~18% total revenue loss due to different contract types.

3.Payment Methods Insights:

- 1.Bank Transfer, Credit Card, and Electronic Check has the highest impact of payment methods on customer tenure Approximately (~0.20M).
- 2.Customers using electronic check shows the higher churn compared to other payment methods, indicating that payment friction may influence customer retention.

4.Internet Service Insights:

- 1.Fiber optic users exhibits the highest churn rate (~42%), suggesting potential dissatisfaction despite being the high-value customers.

5.Add-On Service Insights:

- 1.Customers with device protection have significantly higher retention rate (~85-95%), highlighting the importance of value-added services in the reducing churn.

6.Customer Profile Insight:

- 1.Customers Without partners show a higher churn rate (~33%) compared to partnered customers (~19%), indicating that single users are more likely to discontinue the services.

7.Pricing Insights:

- 1.Customers with higher monthly charges and lower tenure are more likely to churn, suggesting a mismatch between pricing and value.

Business Recommendations

Promote long-term contracts through loyalty rewards and bundled offers. Encourage automated payment methods to reduce friction and improve retention. Enhance fiber optic service quality and customer support experience. Implement personalized pricing and value-added service packages for high-risk customers.

Business Recommendations:

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1. 🎯 Encourage Long-Term Contracts:

Introduce discounts, loyalty rewards, and bundled offers for yearly and multi-year plans to reduce churn from month-to-month customers.

2. 💳 Promote Automated Payment Methods:

Encourage customers to shift from electronic checks to auto-pay methods through cashback offers and payment incentives to improve retention.

3. 🌐 Improve Fiber Optic Service Experience:

Focus on service quality, customer support, and pricing optimization for fiber optic users, as they represent the highest churn-risk customer segment.

4. 📦 Increase Adoption of Value-Added Services:

Bundle device protection and additional support services with internet plans to improve customer retention and long-term engagement.

5. 🏠 Optimize Pricing for High-Charge Customers:

Reassess pricing strategies and provide personalized offers for customers with high monthly charges to reduce dissatisfaction and churn risk.

Conclusion

The analysis reveals that contract structure, payment behavior, service adoption, and pricing are major contributors to customer churn. By focusing on long-term customer engagement, automated payment adoption, service quality improvements, and targeted retention programs, organizations can reduce churn, protect revenue, and improve customer lifetime value.